

The Generation-4 Single-Cell Collagen Model

Theory, Governing Equations, Staged Controls, and Evidence Boundaries

Motor–clutch collagen-remodelling project

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Abstract

Generation 4 (G4) is a two-dimensional, single-cell mechanism-calibration platform in micrometres, nanonewtons, and seconds. It fixes the cell first, calibrates an elastic bead–spring collagen network, traces crosslink-mediated force transmission, adds stochastic motor–clutch failure, and finally releases rigid-cell translation and rotation. G4 v1 is the short-time prototype. G4 v2 extends the observation window and exposes small displacements and fast rupture events without inflating the physical parameters. This document distinguishes implemented equations from modelling assumptions and literature-supported mechanisms. G4 is not yet a calibrated three-dimensional tumour-migration prediction.

KEY EQUATION

$$\zeta_b \dot{\mathbf{r}}_i = \mathbf{F}_i^{\text{stretch}} + \mathbf{F}_i^{\text{bend}} + \mathbf{F}_i^{\text{xlink}} + \mathbf{F}_i^{\text{rep}} + \mathbf{F}_i^{\text{active}}. \quad (1)$$

Equation (1) is the common G4 mechanical backbone: inertia-free collagen beads move according to elastic, contact, and active forces divided by bead drag.

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1 Positioning and stage logic

G4 was introduced because earlier generations mixed changes in network geometry, anchoring, contact, and cell mobility. G4 instead follows a controlled inheritance rule: once a mechanism is accepted, the next stage changes one major ingredient while retaining the previous mechanics.

Stage	New test	Primary interpretation
G4A	One-factor elastic calibration with a fixed cell	Sensitivity of displacement, traction, radial order, and boundary response to one coefficient at a time.

G4B	Crosslink-graph transmission	Whether non-contact fibres move only through retained elastic connections.
G4C	Independent versus shared-load clutch failure	Binding, loading, rupture cascades, complete site loss, traction drop, and fibre recoil.
G4D	Release rigid-cell translation and rotation	Whether asymmetric collagen reactions are sufficient to generate motion without prescribed polarity.

G4 v1 runs for two to five minutes and reveals the controls, but many indirect displacements are subpixel and many clutch events occur between saved frames. G4 v2 retains the same single-cell concept and parameter language while using two-hour elastic and moving-cell runs, a 30-minute clutch overview, and a 20-second event microscope sampled every 0.1 seconds.

2 Collagen network mechanics shared by G4A–D

Each collagen fibre is a connected chain of visible beads. Consecutive beads carry axial force; triplets resist changes in curvature; retained intersections couple material points on different fibres. Only beads in the outer boundary band are anchored in the main condition.

2.1 Overdamped update

For every non-fixed bead,

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \frac{\Delta t}{\zeta_b} \mathbf{F}_i^{\text{total}}, \quad \mathbf{r}_i(t) = \mathbf{r}_i^0 \quad (i \in \mathcal{B}_{\text{fixed}}). \quad (2)$$

Bead drag controls the transient speed. It dissipates motion but stores no elastic energy. The bonds and crosslinks store energy. G4 contains no standard-linear-solid element, inertia, thermal Brownian forcing, or permanent remodelling.

2.2 Axial stretch and compression softening

For a bond from bead i to j ,

$$\begin{aligned} \mathbf{d}_{ij} &= \mathbf{r}_j - \mathbf{r}_i, & \ell_{ij} &= \|\mathbf{d}_{ij}\|, & e_{ij} &= \ell_{ij} - \ell_{ij}^0, & (3) \\ \mathbf{F}_{i \leftarrow j}^{\text{stretch}} &= k_{ij} e_{ij} \hat{\mathbf{e}}_{ij}, & \hat{\mathbf{e}}_{ij} &= \mathbf{d}_{ij} / \ell_{ij}, & k_{ij} &= \begin{cases} EA / \ell_{ij}^0, & e_{ij} \geq 0, \\ \rho EA / \ell_{ij}^0, & e_{ij} < 0. \end{cases} & (4) \end{aligned}$$

The baseline compression ratio is $\rho = 0.10$. Tension therefore uses the full axial stiffness, whereas compression is softened to represent collagen microbuckling.

2.3 Discrete bending

The curvature stencil for an interior triplet is

$$\Delta \kappa_i = (\mathbf{r}_{i-1} - 2\mathbf{r}_i + \mathbf{r}_{i+1}) - \kappa_i^0, \quad U_b = \frac{1}{2} \sum_i \frac{EI}{\bar{\ell}_i^3} \|\Delta \kappa_i\|^2. \quad (5)$$

The reference curvature κ_i^0 is the initial fibre shape. The bending multiplier changes EI without changing the axial rigidity EA .

2.4 Permanent material-point crosslinks

For an intersection located at fractions α_a and α_b along two host bonds,

$$\mathbf{p}_a = (1 - \alpha_a)\mathbf{r}_{a_1} + \alpha_a\mathbf{r}_{a_2}, \quad \mathbf{p}_b = (1 - \alpha_b)\mathbf{r}_{b_1} + \alpha_b\mathbf{r}_{b_2}, \quad (6)$$

$$\mathbf{F}_x = k_x [(\mathbf{p}_b - \mathbf{p}_a) - \mathbf{d}_x^0]. \quad (7)$$

Interpolation weights distribute $\mathbf{F}_x$ back to the four host beads. The link resists separation but is freely hinged: it does not prescribe the angle between fibres. G4 crosslinks do not rupture or re-form.

3 G4A: one-factor mechanics calibration

G4A keeps the cell fixed and compares the same initial collagen geometry while varying one coefficient at a time. The left panel is the baseline control; the right panel is the selected variant. Both use the same cell position, candidate intersections, duration, and random seed.

Candidate intersections receive stored uniform marks u_j . Crosslink probability changes the network through a nested rule:

KEY EQUATION

$$\text{retain link } j \iff u_j < p_x, \quad u_j \sim \mathcal{U}(0, 1), \quad (8)$$

so increasing p_x adds links without reshuffling the links already present at lower p_x .

Coefficient	Equation location	Expected effect when increased
Total pull F_{pull}	Active contact force	More direct loading until stiffness or geometry limits dominate.
Bending multiplier	EI in Eq. (5)	Fibres resist rotation and straightening.
Collagen modulus E	EA and EI	Axial extension and usually local displacement decrease.
Bead drag ζ_b	Eqs. (1)–(2)	Transients slow; the final elastic equilibrium should not change.
Crosslink probability p_x	Eq. (8)	More force paths, with possible additional rotational constraint.
Crosslink stiffness k_x	Eq. (7)	Each retained link transmits separation force more strongly.
Contact width σ_c	Eq. (10)	The same finite pull is spread more evenly among eligible contacts.

The mobile-boundary control releases outer anchors only in the comparison panel. It identifies whole-network translation that might otherwise be mistaken for local remodelling. The G4 v2 elastic overview is two hours with sparse display frames; the solver step remains $\Delta t = 0.05$ s, so display sampling does not change the physics.

4 Hybrid cell–collagen coupling

The closest material point on each candidate bead-to-bead segment is computed first. A point is eligible only when it lies inside a finite surface-contact shell. Gaussian weights are normalized only over this eligible set C :

KEY EQUATION

$$d_m = \|\mathbf{p}_m - \mathbf{r}_c\| - R, \quad 0 \leq d_m \leq w_c, \quad (9)$$

$$w_m = \frac{\exp(-d_m^2/\sigma_c^2)}{\sum_{n \in C} \exp(-d_n^2/\sigma_c^2)}. \quad (10)$$

The baseline uses $w_c = 3 \mu\text{m}$ and $\sigma_c = 1.5 \mu\text{m}$. The Gaussian therefore distributes a finite local pull; it is not evaluated over the whole ECM.

The active force and its interpolation to the two host beads are

$$\mathbf{F}_m^{\text{active}} = F_{\text{pull}} w_m \frac{\mathbf{r}_c - \mathbf{p}_m}{\|\mathbf{r}_c - \mathbf{p}_m\|}, \quad (11)$$

$$(\mathbf{F}_i^{\text{active}}, \mathbf{F}_j^{\text{active}}) = ((1 - \alpha_m)\mathbf{F}_m^{\text{active}}, \alpha_m\mathbf{F}_m^{\text{active}}). \quad (12)$$

Using a material point avoids bead-resolution artefacts and preserves both total force and its first moment along the bond.

Beads cannot penetrate the rigid circular cell:

$$\delta_i = \max(0, R + c - \|\mathbf{r}_i - \mathbf{r}_c\|), \quad \mathbf{F}_i^{\text{rep}} = k_{\text{rep}} \delta_i \frac{\mathbf{r}_i - \mathbf{r}_c}{\|\mathbf{r}_i - \mathbf{r}_c\|}. \quad (13)$$

This one-sided penalty is contact, not adhesion.

5 G4B: crosslink-mediated indirect realignment

Only direct-contact fibres receive Eq. (12). Every non-contact fibre receives zero Gaussian force. Fibres are classified by shortest path through the retained crosslink graph:

KEY EQUATION

$$d_{\text{graph}}(f, C_{\text{direct}}) \in \{0, 1, 2+, \infty\}. \quad (14)$$

Here 0 denotes direct contact, 1 one retained crosslink hop, 2+ a longer connected path, and ∞ an unconnected fibre. An indirect fibre moves because deformation of a direct fibre separates the linked material points, Eq. (7) produces equal-and-opposite force, and the neighbouring fibre redistributes that force through its own stretching and bending.

For nematic orientation, a fibre pointing at θ is equivalent to one pointing at $\theta + \pi$. The implemented orientation summaries are

$$\theta_f = \frac{1}{2} \text{atan2}(\langle \sin 2\theta_e \rangle, \langle \cos 2\theta_e \rangle), \quad (15)$$

$$\Delta\theta_f = \frac{1}{2} \text{atan2}(\sin 2(\theta_f - \theta_f^0), \cos 2(\theta_f - \theta_f^0)), \quad (16)$$

$$u_f = \frac{1}{N_f} \sum_{i \in f} \|\mathbf{r}_i - \mathbf{r}_i^0\|. \quad (17)$$

Each graph class reports displacement and orientation distributions. In the no-crosslink negative control, non-contact transmitted motion should collapse toward the unconnected class.

6 G4C: motor-clutch loading and stochastic slippage

G4C replaces deterministic attachment with effective clutches that bind, load, rupture, and rebind while the cell remains fixed. Actin flow slows with site load, and relative motion loads a Hookean clutch:

KEY EQUATION

$$v_a = v_0 \max\left(0, 1 - \frac{F_{\text{site}}}{F_{\text{stall}}}\right), \quad \dot{x} = \max(0, v_a - v_{\text{sub}}), \quad f = k_c x. \quad (18)$$

The Bell slip-bond law gives a continuous force-dependent hazard rather than a deterministic “too long” threshold:

$$k_{\text{off}}(f) = k_{\text{off}}^0 \exp\left(\frac{f}{F_b}\right), \quad P_{\text{off}} = 1 - \exp[-k_{\text{off}}(f)\Delta t], \quad P_{\text{on}} = 1 - \exp(-k_{\text{on}}\Delta t). \quad (19)$$

In the independent baseline,

$$F_{\text{site}} = \sum_{j \in \text{bound}} k_c x_j. \quad (20)$$

One rupture removes only that clutch. The whole attachment persists while at least one clutch is bound.

In the shared-load alternative, if i of N clutches remain bound,

KEY EQUATION

$$f_i = \frac{F_{\text{site}}}{i}, \quad r_i = i k_{\text{off}}^0 \exp\left(\frac{F_{\text{site}}}{i F_b}\right), \quad g_i = (N - i) k_{\text{on}}. \quad (21)$$

After a rupture, fewer survivors carry the site load, increasing force per survivor and the Bell off-rate. Rebinding may rescue the site. Complete failure is the first passage to $i = 0$. The implementation also assumes

$$K_{\text{site}} = N k_c \quad (i > 0), \quad F_{\text{site}} = K_{\text{site}} x_{\text{site}}, \quad (22)$$

which holds the fully-bound bundle stiffness until the site is completely lost. This is an explicit coarse-grained closure, not a molecular identity.

The effective defaults are $N = 12$, $k_c = 2 \text{ nN}/\mu\text{m}$, $k_{\text{on}} = 0.055 \text{ s}^{-1}$, $k_{\text{off}}^0 = 0.018 \text{ s}^{-1}$, $F_b = 1.5 \text{ nN}$, $v_0 = 0.025 \mu\text{m/s}$, and $F_{\text{stall}} = 8 \text{ nN/site}$. They describe effective bundles, not twelve counted integrin molecules.

7 G4D: reaction-driven rigid-cell motion

G4D retains the G4C shared-load mechanics and releases only cell translation and rotation.

KEY EQUATION

$$\gamma_c \dot{\mathbf{r}}_c = - \sum_m \mathbf{F}_m^{\text{cell} \rightarrow \text{ECM}}. \quad (23)$$

The cell receives the equal-and-opposite reaction to the traction it applies to collagen. No $+x$ force, preferred velocity, or hardcoded polarity probability is supplied. Rotation obeys

$$\gamma_\theta \dot{\phi} = \sum_m (\mathbf{a}_m - \mathbf{r}_c) \times [-\mathbf{F}_m^{\text{cell} \rightarrow \text{ECM}}], \quad \gamma_\theta = \chi \gamma_c R^2. \quad (24)$$

Heterogeneous collagen geometry, different force paths, and stochastic site lifetimes create a spatial traction imbalance. The fixed and moving panels share the network and random proposals; only rigid-body degrees of freedom differ. Contacts are periodically refreshed as the cell and network move; a bound site retains its material attachment until release.

8 How to read the visualization

Collagen is shown as coloured curves with small beads connected by axial bonds. Orange marks the direct-contact graph class, blue one-hop fibres, and purple two-plus-hop fibres. Gold diamonds or short connectors are permanent elastic crosslinks. Clutch spokes are bound effective clutches; red marks are clutch rupture events, not fibre or crosslink breaking.

Thin green lines are displacement vectors, not collagen, force links, or crosslinks:

KEY EQUATION

$$\Delta \mathbf{r}_i = \mathbf{r}_i(t) - \mathbf{r}_i(0), \quad \mathbf{r}_i^{\text{display}} = \mathbf{r}_i(0) + M \Delta \mathbf{r}_i, \quad M \in \{1, 10, 50\}. \quad (25)$$

Only the green vector length is magnified. True bead and fibre positions remain at $1\times$. One-hop mean displacement may be only 10^{-3} – $10^{-2} \mu\text{m}$ in a $180 \mu\text{m}$ full-field view, so local zoom and numerical readouts are required.

9 Parameter provenance and calibration boundaries

Parameter	Baseline or range	Interpretation status
Domain / fibres	$180 \mu\text{m} / 99$	Inherited G2-scale fixture; not image-derived density.
Cell radius	$10 \mu\text{m}$	Single-cell fixture; replace with experiment-specific segmentation.
Fibre length / diameter	$20\text{--}80 \mu\text{m} / 0.30 \mu\text{m}$	Effective fibre scale, not a molecular fibril.
Collagen modulus	3 MPa	Mechanism-first choice, not a tissue-matched fit.
Bending multiplier	$0.10, 0.25, 0.50, 1.0$	Sensitivity separated from axial stiffness.
Crosslink probability	$0, 0.20, 0.35, 0.60$	Topology sensitivity; a 2-D crossing is not automatically a biological weld.
Crosslink stiffness	$5, 10, 25, 75 \text{ nN}/\mu\text{m}$	Effective link-compliance sensitivity.
Bead drag	$90, 180, 360 \text{ nN s}/\mu\text{m}$	Transient sensitivity; requires time-resolved fitting.
Total pull	$12, 24, 48 \text{ nN}$	Mechanism range; not fitted cell traction.
Contact shell / σ_c	$3 \mu\text{m} / 1.5 \mu\text{m}$	Hybrid local-contact starting choice.
Cell drag	$600 \text{ nN s}/\mu\text{m}$	Effective mobility; requires ensemble calibration.

10 Verification gates, limitations, and references

The required verification gates are: time-step convergence between 0.05 and 0.025 s; zero traction with no bound clutch; collapse of non-contact transmitted motion when crosslinks are removed; traceable one-hop force paths; increasing

shared-load hazard as i decreases at fixed site load; complete site loss only at $i = 0$; agreement among rupture logs, traction drops, and animation frames; and paired fixed/moving runs that differ only in cell degrees of freedom.

G4 is limited to a 2-D projection, one rigid circular cell, linear permanent crosslinks, and mechanism-first effective parameters. It does not include a nucleus, deformable protrusions, three-dimensional pores, crosslink turnover, fibre sliding, damage, strain stiffening, SLS relaxation, proteolysis, proliferation, or biochemical polarity.

Key references

1. Lee et al. (2014), *PLOS ONE* 9:e111896. doi:10.1371/journal.pone.0111896.
2. Abhilash et al. (2014), *Biophysical Journal* 107:1829–1840. doi:10.1016/j.bpj.2014.08.029.
3. Wang et al. (2014), *Biophysical Journal* 107:2592–2603. doi:10.1016/j.bpj.2014.09.044.
4. Notbohm et al. (2015), *Journal of the Royal Society Interface*. arXiv:1407.3510.
5. Bell (1978), *Science* 200:618–627. doi:10.1126/science.347575.
6. Bangasser and Odde (2013), *Cellular and Molecular Bioengineering* 6:449–459.
7. Riching et al. (2014), *Biophysical Journal* 107:2546–2558. PMCID: PMC4255204.
8. Erdmann and Schwarz (2004), arXiv:cond-mat/0403552.
9. Gao, Qian, and Chen (2011), *Journal of the Royal Society Interface* 8:1217–1232.

Claim boundary. G4 can separate elastic parameter sensitivity, crosslink-mediated indirect motion, clutch failure, and reaction-driven rigid-cell motion. It cannot yet predict whole-tumour invasion.